PySpark Operations

Friday, 4 September 2026 6:16 PM

* Transformation vs Action.

Aspect	Transformation	Action.
Purpose	Defines how data should be processed	Triggers execution and produces result
Execution	Lazy (Not executed immediately)	Executed immediately.
Effect on data	Returns a new Dataframe or RDD	Returns a value to written to driver or disk
Optimization	Can be combined or reordered by Spark.	Cannot be optimized once triggered.
Resource usage	No cluster resources used at definition	Uses cluster resources.

*

customers_df =

id	first_name	last_name	mobile	city
1	A	W	111	I
2	B	X	222	Null
3	"Unknown"	Y	333	K
4	D	Null	444	Null

⟹ Data scientist

subset = [first_name, last_name].

Customer	Order_value	is_churn	is_active
A	100	0	N
A	200	1	Y
B	300	0	N
C	200	1	Y
A	500	1	Y
B			

count = 5

Sum = 3.

Customer	OV
A	100
A	200

Group1 → Sum →

Customer	OV
A	300

Customer	OV
B	200
B	500

Group2 →

Customer	OV
B	700

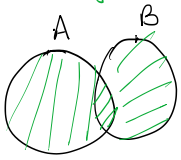
Customer	OV
C	300

Group3 →

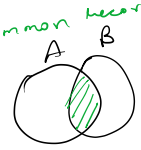
Customer	OV
C	300

* Joins:

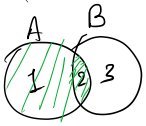
1) Outer: All records from both datasets.



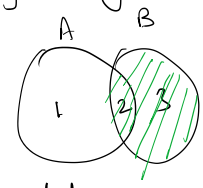
2) Inner: Common records.



3) Left: All from left, matching from right.



4) Right: All from right and matching from left.



Select, as a
t1.a, t2.b as b.

inner join.

table1 t1

table2 t2

on t1.key = t2.key.

a | b

* Accumulator vs Broadcast variables.

⇒ Accumulator: - Shared variable in pyspark.
- Used for counting, summing or tracking.
- Aggregation allowed across multiple tasks.

⇒ Broadcast variable. - Shared variable.
- Read only.